THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A student tests the effect of the mass of a bob on the time period of a pendulum. Which is the dependent variable? 1

(a) length of the string **(b)** mass of the bob
(c) place of testing **(d)** time period

2. A microscope has a 10x eyepiece and a 40x objective. A cell appears 2 mm wide in the image. The actual width is: 1

(a) 800 mm **(b)** 0.5 mm **(c)** 0.005 mm **(d)** 0.05 mm

3. A person has a fever, a rash and a headache. Which of these is a sign, not a symptom? 1

(a) body ache **(b)** feeling weak
(c) headache **(d)** rash seen by the doctor

4. A student connects a bulb, a switch and a cell but the bulb does not glow even when the switch is closed. The bulb holder is loose. The fault is a: 1

(a) break in the circuit **(b)** magnetic effect
(c) overload **(d)** short circuit

5. After heating, the temperature readings are 40, 60, 80 and 100 degrees C at times 1, 2, 3 and 4 min. The rate of rise is: 1

(a) 40 degrees C per minute **(b)** 100 degrees C per minute
(c) 20 degrees C per minute **(d)** 10 degrees C per minute

6. A tissue differs from an organ because a tissue: 1

(a) has similar cells doing one job **(b)** has many types of tissues
(c) forms systems **(d)** is an organism

7. The correct sequence of events in infection by a pathogen is: 1

(a) symptoms, entry, multiplication **(b)** symptoms, multiplication, entry
(c) multiplication, symptoms, entry **(d)** entry, multiplication, symptoms

8. A toaster, an iron and a heater use the same effect of current, which is: 1

(a) heating **(b)** light **(c)** magnetic **(d)** chemical

9. A data table has more than one outlier. The best action is to: 1

(a) check the experimental method **(b)** use the biggest
(c) drop all of them **(d)** stop recording

10. A cell is stained in blue and the nucleus appears dark. The role of stain is to: 1

(a) kill the cell **(b)** increase size **(c)** make food **(d)** provide contrast

11. The immune system includes: 1

(a) teeth and hair **(b)** only skin colour

(c) white blood cells and antibodies **(d)** bones and nails

12. The magnetic effect of current is shown by a: 1

(a) wire melting **(b)** bulb glow

(c) compass needle deflection **(d)** bubble formation

13. In a pendulum test of the effect of the mass of the bob on the time period, which must be kept constant? 1

(a) length of the string **(b)** time period

(c) number of masses **(d)** mass of the bob

14. A cell that has a nucleus, a cell wall and chloroplasts is likely from: 1

(a) a leaf **(b)** human cheek **(c)** bacterium **(d)** blood

15. In a village, many people fell ill with diarrhoea after a flood. The most probable cause is: 1

(a) mosquito bite **(b)** contaminated drinking water

(c) dust **(d)** sunlight

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are the independent and dependent variables different in an investigation? 2

17. Why are bacteria placed in a different group from plants and animals? 2

18. Why is stagnant water a health risk? Name two diseases that result. 2

19. A bulb does not glow in a circuit. List two possible faults and how you would check them. 2

20. State two limitations of using only the senses for observation. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A student plans to test whether sugar dissolves faster in warm water. List the variable to change, the variable to measure and three things to keep same. 3

22. A cell shows a thick wall, large vacuole and green bodies. Identify the cell, name the green bodies and state their function. 3

23. Explain how a vaccine produces immunity. Why are booster doses needed? 3

24. A student made an electromagnet but it can't pick up pins. List three possible reasons and remedies. 3

OR

A student measured the length of a table with a hand span and found 8; another got 9. Explain the difference and suggest a better method. 3

25. A cell is magnified 200x and measures 4 mm in the picture. Find its actual size in micrometres. 3

OR

Differentiate between first aid and medical treatment. Give an example of first aid. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Plan an investigation to find out which of three liquids (water, milk, oil) cools fastest. Mention what you measure, what you keep constant, and the procedure. 5

OR

Draw up a comparison between the plant cell, animal cell and bacterial cell in a table in terms of nucleus, cell wall, chloroplast, vacuole and size. 5

27. Explain the chain of infection of a communicable disease using cholera as an example, and show where it can be broken. 5

OR

Describe an activity to show that a current-carrying wire behaves like a magnet. Mention the observations when the current is reversed and when the compass is placed in different positions. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND